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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SALVACION USA, INC.,
Petitioner,

v.

TRUTEK CORP,
Patent Owner.

IPR2024-00711
Patent 8,163,802 B2

Before ULRIKE W. JENKS, SHERIDAN K. SNEDDEN, and
SUSAN L. C. MITCHELL, *Administrative Patent Judges.*

SNEDDEN, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)
Dismissing Petitioner's Motion to Exclude
37 C.F.R. § 42.64(c)
Dismissing Petitioner's Motion to Strike
37 C.F.R. § 42.23

I. INTRODUCTION

Salvacion USA, Inc. ("Petitioner") filed a Petition requesting an *inter partes* review of claims 1–3 and 8 of U.S. Patent No. 8,163,802 B2 (Ex. 1002, "the '802 patent"). Paper 2 ("Pet."). Trutek Corp. ("Patent Owner") filed a Preliminary Response to the Petition. Paper 6.

We instituted trial on December 6, 2024. Paper 9 ("Inst. Dec."). During trial, Patent Owner filed a Patent Owner Response. Paper 6 ("PO Resp."). Petitioner filed a Reply (Paper 12 ("Pet. Reply")) and Patent Owner filed a Sur-reply (Paper 13 ("PO Sur-Reply")). We held an oral hearing on September 8, 2025, and a transcript of that hearing is of record. Paper 31 ("Tr.").

Petitioner also filed a Motion to Exclude (Paper 22), to which Patent Owner filed an Opposition (Paper 23), and Petitioner filed a Reply (Paper 24). We address Petitioner's Motion in Section IV below.

Petitioner's Motion to Strike (Paper 19) remains pending. We address that motion in Section V below.

We have jurisdiction under 35 U.S.C. § 6(b). After considering the full record developed through trial, we determine that Petitioner has proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

II. BACKGROUND

A. Real Parties in Interest

Petitioner identifies itself, Pharmhealth Technologies, Inc., and Salvacion Co. Ltd. as the real parties-in-interest. Pet. 1. Patent Owner identifies itself as the real party-in-interest. Paper 5, 1.

B. Related Matters

The parties relay that Patent Owner has asserted the ’802 patent against Petitioner in *Trutek Corp. v. Cho, et al*., Case No. 2:23cv3709 (D.N.J. 2023); *Trutek Corp. v. Bluewillow Biologics, Inc*., Case No. 2:21cv10312 (E.D. Mich. 2021); and *Matrixx Initiatives, Inc. v. Trutek Corp*., Case No. IPR2020-01592 (PTAB Sept. 4, 2020). Pet. 1; Paper 5, 1–2. Patent Owner also lists as related matters *Trutek Corp. v. Matrixx Initiatives, Inc., et al.*, Case No. 3:19cv17647 (D.N.J. 2019) and *Trutek Corp. v. Bluewillow Biologics, Inc*., Appeal No. 24-1555 (Fed. Cir. 2024). Paper 5, 1.

C. The ’802 patent (Ex. 1001)

The ’802 patent is directed to electrostatically charged nasal application products that reduce the inflow of airborne contaminants to the nasal passages, by capturing the contaminants, such as viruses, bacterial, and other harmful microorganisms or toxic particulates, and keeping them from entering the body, while also inactivating them to render them harmless. Ex. 1001, 1:67–2:7. According to the Specification,

> [t]his formulation, when applied creates an electrostatic field having a charge. The electrostatic field attracts airborne particulates of opposite charge to the substrate that are in close proximity to the substrate close to the skin and a biocidic agent renders microorganisms coming in contact [with] the substrate or skin less harmful.

Id. at Abst. That is, the ’802 patent discloses methods and compositions capable of capturing particulates and microorganisms for sufficient duration so as to “inactivate, kill, or render harmless a microorganism.” *Id.* at 2:62–3:8.

More specifically, the ’802 patent discloses electrostatically charged compositions comprising polymeric quaternary compounds, such as polyquaternary ammonium compounds, that are “capable of creating an electrostatic field on and around a surface to which it is applied.” *Id.* at 3:35–40, 4:39–46. “[W]hen applied to a surface, [the polymeric quaternary compound] creates an electrostatic field such that oppositely charged airborne particulates (including microorganisms) in the vicinity of the surface are electrostatically trapped, held thereto and one or more of the microorganisms so captured is neutralized, killed, inactivated, and rendered harmless.” *Id.* at 3:35–40.

The ’802 patent provides exemplary formulations which comprise a quaternary thickener, e.g., a polyquaternary ammonium compound, and a biocidic agent. *Id*. at 5:55–10:5. The ’802 patent also explains that benzalkonium chloride may also serve the same function as a quaternary thickener, but that it is also a cationic agent and a biocide. *Id*. at 5:25–26.

D. The Challenged Claims

Challenged claims 1–3 and 8 of the ’802 patent are set forth below.

> 1. A method for electrostatically inhibiting harmful particulate matter from infecting an individual through nasal inhalation wherein a formulation is applied to skin or tissue of nasal passages of the individual in a thin film, said method comprising:
>
> > a) electrostatically attracting the particulate matter to the thin film;
> >
> > b) holding the particulate matter in place by adjusting the adhesion of the thin film to permit said thin film to stick to the skin or tissue and by adjusting the cohesion of the formulation to provide adequate impermeability to the thin film; and,

> c) inactivating the particulate matter by adding at least one ingredient that would render said particulate matter harmless.
>
> 2. A formulation for electrostatically inhibiting harmful particulate matter from infecting an individual through nasal inhalation wherein the formulation is applied to skin or tissue of nasal passages of the individual in a thin film, said formulation comprising at least one cationic agent and at least one biocidic agent, and wherein said formulation, once applied:
>
> a) electrostatically attracts the particulate matter to the thin film;
>
> b) holds the particulate matter in place by adjusting the adhesion of the thin film to permit said thin film to stick to the skin or tissue and by adjusting the cohesion of the formulation to provide adequate impermeability to the thin film; and,
>
> c) inactivates the particulate matter and renders said particulate matter harmless.
>
> 3. The formulation of claim 2 wherein the at least one cationic agent is a polymeric quaternary ammonium compound.
>
> 8. A formulation for electrostatically inhibiting harmful particulate matter from infecting an individual through nasal inhalation wherein the formulation is applied to skin or tissue of nasal passages of the individual in a thin film, said formulation comprising:
>
> a) at least one biocidic agent, and
>
> b) at least one quaternary thickener.

Ex. 1001, 10:19–51; 11:1–7.

The parties refer to claim elements 1a and 2a as the "CATCH" function. *See*, *e.g*., PO Resp. 10; Pet. Reply 8. The parties refer to claim elements 1b and 2b as the "HOLD" function. *See*, *e.g*., PO Resp. 11; Pet. Reply 11. The parties refer to claim elements 1c and 2c as the "KILL" function. *See*, *e.g*., PO Resp. 8. That is, "the terms CATCH, HOLD, and KILL are mere abbreviations of the functions of Elements (a), (b), and (c), respectively." *See*, *e.g*., PO Sur-Reply 5–6, 11 n.11.

E. Evidence

Petitioner relies upon information that includes the disclosures of the following references:

> Ex. 1005, Chen, U.S. Patent App. Pub. 2005/0232895 A1, published October 20, 2005 ("Chen").
>
> Ex. 1006, Baker, Jr. et al., U.S. Patent 6,559,189 B2, issued May 6, 2003 ("Baker").
>
> Ex. 1007, Wahi et al., U.S. Patent 6,844,005 B2, issued January 18, 2005 ("Wahi '005").
>
> Ex. 1008, Rolf, U.S. Patent App. Pub. 2004/0071757 A1, published April 15, 2004 ("Rolf").

Petitioner also relies upon the Declaration of Kinam Park, Ph.D. (Ex. 1002).

F. Asserted Grounds of Unpatentability

Petitioner asserts that claims 1–3 and 8 are unpatentable on the following grounds (Pet. 13):

Ground	Claims Challenged	35 U.S.C. §[1]	References
1	1–3, 8	102(b)	Chen
2	1–3, 8	103	Chen
3	1–3, 8	102(b)	Baker
4	1–3, 8	103	Baker
5	1–3, 8	102(b)	Wahi ’005
6	1–3, 8	103	Wahi ’005
7	1, 2, 8	102(b)	Rolf
8	1, 2, 8	103	Rolf
9	2	103	Rolf, Wahi ’005

Pet. 13.

G. Claim Construction

The Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b); 37 C.F.R. § 100(b). Under that standard, claim terms “are generally given their ordinary and customary meaning” as understood by a person of ordinary skill in the art ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp*., 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc) (quoting *Vitronics Corp. v. Conceptronic, Inc*., 90 F.3d 1576, 1582 (Fed.

[1] The Leahy-Smith America Invents Act, Pub. L. No. 112–29, 125 Stat. 284 (2011) (“AIA”), amended 35 U.S.C. § 102(b) and §103, effective March 16, 2013. The earliest claimed priority date for the ’802 patent is before the effective date of the applicable AIA amendment. *See* Ex. 1002, (60). Accordingly, we apply the pre-AIA versions of § 102(b) and §103.

Cir. 1996)). "In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence." *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing Phillips, 415 F.3d at 1312–17).

Petitioner proposes constructions for a number of claim terms. Pet. 14–16. We address those proposals to the extent that express construction of the term is necessary for purposes of resolving the dispute between the parties. *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (Only those terms that are in controversy need be construed, "and only to the extent necessary to resolve the controversy."). Insofar as the parties dispute the mapping of the prior art to the challenged claims, we address that dispute below including, to the extent necessary, any implicit disagreement about the subject matter encompassed by the challenged claims.

1. Limiting Effect of the Preamble of Independent Claims 1, 2, and 8

In our Institution Decision, we preliminarily construed the preamble of independent claims 1, 2, and 8 as limiting. Inst. Dec. 7–8. The parties do not dispute our construction of the preambles of claims 1, 2, and 8 as limiting. *See generally* PO Resp.; Pet. Reply 2–4. After consideration of the full trial record, we maintain our construction of the preamble of claims 1, 2, and 8 as limiting for the reasons set forth in our Institution Decision. *Id.* Briefly, the preamble of independent claims 1, 2, and 8 limits the challenged claims as both the preamble and the body are used to define the subject matter claimed. *See Allen Engineering Corp., v. Bartell Industries, Inc.*, 299 F.3d 1336, 1346 (Fed. Cir. 2002). In particular, the preamble is "necessary to give life, meaning and vitality" to the claim, *see id*., because it provides the only description in the claim for the application site of the formulation, i.e., "to skin or tissue of nasal passages," and how the formulation is applied, i.e., "in a thin film."

2. *"electrostatically inhibiting" and "electrostatically attracting/attracts"*

The preamble of independent claims 1, 2, and 8 recite the term "electrostatically inhibiting." Claim 1 additionally recites that the formulation is capable of "electrostatically attracting" the particulate matter to the thin film, while claim 2 recites that the formulation "electrostatically attracts" the same. In our Institution Decision, we preliminarily construed "electrostatically inhibiting" to refer to the claimed process of using an electrostatic field to attract the charged particulate matter and then inactivating the matter to render it harmless. Inst. Dec. 11–12. The parties do not oppose our construction. *See generally* PO Resp.; Pet. Reply 2–4.

After consideration of the full trial record, we maintain our construction. From our perspective, the claims themselves support this construction as the claims expand upon the term "electrostatically inhibiting" with specific means for achieving such inhibition. Specifically, claims 1 and 2 recite electrostatically attracting the particulate matter, holding the matter in place, and inactivating the matter. Additionally, the Specification supports our construction as it describes the use of "an electrostatic field such that oppositely charged airborne particulates (including microorganisms) in the vicinity of the surface are electrostatically trapped, held thereto and one or more of the microorganisms so captured is neutralized, killed, inactivated, and rendered harmless." Ex. 1001, 3:36–40; PO Resp. 10 n.6.

3. "adequate impermeability"

Independent claims 1 and 2 recite "adjusting the cohesion of the formulation to provide adequate impermeability to the thin film." The term "adequate impermeability" occurs only in the claims of the '802 patent.

We adopt the parties' construction of "adequate impermeability" that has been accepted in the related district court litigation:

> "Adequate impermeability" refers to the "thin film holding harmful particles in place and inhibiting them from penetrating the thin film and contacting the skin or tissue of an individual's nasal passages. This is done by varying the concentration of ingredients . . .[,] thereby adjusting the adhesion and cohesion of the thin film."

Pet. Reply 3 (quoting Ex. 1010, 17). Furthermore, as stated in our Institution Decision, along with that term, the claim language explains that the thin film is meant to stick to the skin or tissue and to hold the particulate matter in place. Inst. Dec. 9–10. The claim language also explains that the cohesion of the formulation is adjusted to provide adequate impermeability to the thin film. Thus, it is reasonable to consider the recited impermeability is a function of the cohesive properties of the thin film, and that such impermeability refers to a property of the thin film, i.e., inhibiting the penetration or passage of matter through it onto the skin or tissue at the application site.

4. "adhesion" and "cohesion"

Claims 1 and 2 recite "adjusting the adhesion of the thin film to permit said thin film to stick to the skin or tissue" and "adjusting the cohesion of the formulation to provide adequate impermeability to the thin film." The terms "adhesion" and "cohesion" appear only in the claims of the '802 patent.

In our Institution Decision, we determined that the record supports finding that a skilled artisan would understand the ordinary and customary meaning of the term "adhesion" is the tendency of dissimilar substances to adhere or stick together. Inst. Dec. 12–13. The parties do not oppose our construction. *See generally* PO Resp.; Pet. Reply 2–4. Similarly, the skilled artisan would understand the ordinary and customary meaning of the term "cohesion" is the tendency of similar or identical substances to adhere or stick together. In that context, the claim language provides support for these constructions because the claims explain that the adhesion of thin film permits it to "stick" to the skin or tissue and that the cohesion of the formulation provides the adequate impermeability to the thin film. After consideration of the full trial record, we maintain our construction.

H. Level of Ordinary Skill in the Art

The level of skill in the art is a factual determination that provides a primary guarantee of objectivity in an obviousness analysis. *Al-Site Corp. v. VSI Int'l Inc.*, 174 F.3d 1308, 1323 (Fed. Cir. 1999) (citing *Graham v. John Deere Co*., 383 U.S. 1, 17–18 (1966)); *Ryko Mfg. Co. v. Nu-Star, Inc*., 950 F.2d 714, 718 (Fed. Cir. 1991)).

In our Institution Decision, we adopted Petitioner's definition of a person of ordinary skill in the art ("POSITA") as it was undisputed at the time and consistent with the evidence of record. Inst. Dec. 7–8. Based on our consideration of the full record, we maintain our determination that a POSITA at the time the application for the '802 patent was filed "would have had at least an undergraduate degree in pharmacy, chemistry, biochemistry, biomedical engineering or a closely related field, such that the POSITA generally understand[s] pharmaceutical formulation." Pet. 12. Additionally, we agree with Petitioner that "[a] POSITA would be familiar with the chemical ingredients in the ['802] Patent, biocidic agents, and cationic agents, have knowledge of airborne particles including viruses and bacteria, and understand electrostatic fields, electrostatic attraction and repulsion, adhesion and cohesion." *Id.* (citing Ex. 1002 ¶¶ 38–39).

We have reviewed the credentials of Petitioner's declarant, Dr. Park, and consider him qualified to provide his opinion on the level of skill and the knowledge of a POSITA at the time of the invention. *See* Ex. 1002 ¶¶ 3–13.

III. ANALYSIS

A. Introduction

"In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). Moreover, a petitioner should not "place the burden on [the Board] to sift through information presented by the Petitioners, determine where each element [of the challenged claims] is found in [the cited references], and identify any differences between the claimed subject matter and the teachings of [the cited references.]" *Google*

Inc. v. EveryMD.com LLC, IPR2014-00347, Paper 9 at 25 (PTAB May 22, 2014).

Anticipation is a question of fact, as is the question of what a prior art reference teaches. *In re NTP, Inc.*, 654 F.3d 1279, 1297 (Fed. Cir. 2011). "Because the hallmark of anticipation is prior invention, the prior art reference—in order to anticipate under 35 U.S.C. § 102—must not only disclose all elements of the claim within the four corners of the document, but must also disclose those elements 'arranged as in the claim.'" *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008) (quoting *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983)). Whether a reference anticipates a claim is assessed from the skilled artisan's perspective. *See Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358, 1368 (Fed. Cir. 2003) ("[T]he 'dispositive question regarding anticipation [i]s whether one skilled in the art would reasonably understand or infer from the [prior art reference's] teaching' that every claim element was disclosed in that single reference." (quoting *In re Baxter Travenol Labs.*, 952 F.2d 388, 390 (Fed. Cir. 1991))).

The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness.[2] *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

[2] Patent Owner does not present any objective evidence of nonobviousness (i.e., secondary considerations) for the challenged claims. *See generally* PO Resp.

The obviousness inquiry also typically requires an analysis of "whether there was an apparent reason to combine the known elements in the fashion claimed by the patent at issue." *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2007) (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006) (requiring "articulated reasoning with some rational underpinning to support the legal conclusion of obviousness")). A petitioner cannot prove obviousness with "mere conclusory statements." *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1380 (Fed. Cir. 2016). Rather, a petitioner must articulate a sufficient reason why a person of ordinary skill in the art would have combined the prior art references. *In re NuVasive*, 842 F.3d 1376, 1382 (Fed. Cir. 2016).

We analyze the asserted grounds of unpatentability in accordance with these principles to determine whether Petitioner has met its burden to establish a reasonable likelihood of success at trial.

B. Grounds 1 and 2: Asserted Unpatentability in view of Chen

Petitioner asserts that Chen anticipates claims 1–3 and 8. Pet. 16–28. To support its contention, Petitioner provides a detailed claim analysis addressing how each element of 1–3 and 8 is disclosed by Chen. *Id.* Petitioner supports this interpretation of Chen with Dr. Park's testimony. Ex. 1002 ¶¶ 108–186. Alternatively, Petitioner asserts that "[i]n the event that one or more of claims 1–3 or 8 are not found to be anticipated by Chen, claims 1-3 and 8 would nevertheless have been obvious over Chen." Pet. 16.

Claims 1, 2, and 8 of the'802 patent are each independent. Claim 1 is directed to a method for electrostatically inhibiting harmful particulate matter from infecting an individual through nasal inhalation by applying a formulation to the skin or tissue of nasal passages. Claims 2 and 8 are directed to formulations for electrostatically inhibiting harmful particulate matter from infecting an individual through nasal inhalation where the formulation comprises at least one cationic agent and at least one biocidic agent. Claim 8 specifies that the claimed formulation comprises at least one biocidic agent and at least one quaternary thickener.

Claim 1 is identical to claim 2, except that claim 1 recites a method for electrostatically inhibiting harmful particular matter using a formulation, without reciting the components of the formulation, whereas claim 2 recites a formulation for such inhibition comprising at least one cationic agent and at least one biocidic agent. Pet. 23; *compare* claim 1, *with* claim 2. Like Petitioner, our analysis focuses on independent claim 2 to determine whether claims 1 and 2 are anticipated by Chen. *See*, *e.g.*, Pet. 23–26 (relying substantially on analysis of claim 2 for independent claim 1).

1. Independent Claim 2

a) Preamble: "A formulation for electrostatically inhibiting harmful particulate matter from infecting an individual through nasal inhalation wherein the formulation is applied to skin or tissue of nasal passages of the individual in a thin film, said formulation comprising at least one cationic agent and at least one biocidic agent, and wherein said formulation, once applied"

Petitioner asserts that Chen teaches the preamble in claim 2, i.e., "[a] formulation for electrostatically inhibiting harmful particulate matter from infecting an individual through nasal inhalation," by disclosing an antiviral composition that is administered to the skin or nasal membrane. *Id*. at 17 (citing Ex. 1005 ¶ 44 (disclosing antiviral compositions administered to skin, oral membrane, or nasal membrane)). Additionally, Petitioner asserts that

Chen discloses that its composition comprises a cationic polymer. *Id*. (citing Ex. 1005 ¶ 78). According to Petitioner's declarant, Dr. Park, such a composition would function to electrostatically inhibit harmful particulate matter from infecting an individual through nasal inhalation. *Id.* (citing Ex. 1002 ¶ 112).

Petitioner also asserts that Chen discloses that its formulation is applied to the skin or tissue of nasal passages in a thin film, as required by claim 2, because Chen explicitly teaches administering the composition locally, such as to skin or to nasal membrane. *Id*. (citing Ex. 1005 ¶ 44). Additionally, Petitioner asserts that Chen describes its formulations as providing "mucoadhesiveness by forming a physical barrier over the mucous membrane." *Id.* at 18 (citing Ex. 1005 ¶¶ 121, 148). Petitioner also asserts that Chen discloses that its compositions may be applied locally as a paste, gel, lotion, cream, or ointment. *Id.* (citing Ex. 1005 ¶ 111). According to Dr. Park, a skilled artisan would understand that such dosage forms would be applied in a thin film. *Id*. (citing Ex. 1002 ¶ 115).

Additionally, Petitioner asserts that Chen discloses that its formulation comprises "at least one cationic agent," as required by claim 2, by disclosing the use of cationic polymers, including poly(methyl methacrylate-b-Nmethyl-4-vinyl pyridinium iodide), which Dr. Park explains is a polymeric quaternary ammonium compound. Ex. 1005 ¶ 78; Ex. 1002 ¶ 117. Petitioner asserts that Chen also discloses that its formulation comprises "at least one biocidic agent," as required by claim 2, by teaching the use of benzalkonium chloride (BAC), which is the same biocidic agent recognized in the '802 patent. *Id*. at 18–19 (citing Ex. 1005 ¶ 86, Ex. 1001, claims 7, 9).

Patent Owner does not raise any arguments regarding the preamble of claim 2, and acknowledges that "Chen's compositions comprise at least one biocidic agent." *See*, *e.g.*, PO Resp. 9–11. To the extent that the preamble is limiting, which we have found, we agree with Petitioner that Chen discloses the preamble.

b) Claim Element 2a: "electrostatically attracts the particulate matter to the thin film"

Petitioner asserts that Chen discloses the "CATCH" element of claim 2.[3] Pet. 19–20. Here, Petitioner relies on Dr. Park's testimony that Chen's formulation, "once applied[,] electrostatically attracts the particulate matter to the thin film," as required by claim 2, because Chen's cationic agents, particularly the quaternary ammonium compound, necessarily electrostatically attracts particulate matter to the topical composition applied as a thin film. *Id*. (citing Ex. 1002 ¶ 127).

Patent Owner disputes Petitioner's contentions that Chen discloses the "CATCH" function of the challenged claims. PO Resp. 9–17. First, Patent Owner contends that "Chen did not anticipate electrostatically attracting airborne virus particles to a person's nostrils." *Id.* at 9. Patent Owner acknowledges that, while "it stands to reason that Chen's positively charged composition would inherently attract negatively charged viruses," "there is no indication that there is a purposeful adjustment of the positive charge so as to effectively create an electrostatic field surrounding the user's nose and

[3] Claims 1 and 2 recite a formulation that "electrostatically attracts the particulate matter to the thin film." As noted above, the parties refer to this element of claims 1 and 2 as the "CATCH" function. *See*, *e.g.*, PO Resp. 10; Pet. Reply 8.

face that would attract airborne particles that float in the vicinity of the nostrils." *Id.* at 9–10.

Having considered the parties' positions and evidence of record, summarized above, we determine that Chen inherently teaches a formulation that "electrostatically attracts the particulate matter to the thin film," the "CATCH" element. Specifically, Chen discloses positively charged antiviral compositions (Ex. 1005 ¶¶ 55–56) that, as acknowledged by Patent Owner, would "inherently attract negatively charged viruses." PO Resp. 9–10. Additionally, we credit Dr. Park's testimony that a skilled artisan at the time of the invention would have understood the cationic polymer in Chen's formulation necessarily electrostatically attracts particulate matter. Ex. 1002 ¶¶ 117, 127. That is all that is required by the "CATCH" element of the claims; "purposeful adjustment of the positive charge" is not a requirement of the claims.

We are not persuaded by Patent Owner's argument that, without the purposeful adjustment of the concentration of cationic ingredients, Chen does not disclose a composition that would achieve the purpose of "electrostatically inhibiting" harmful particulate matter from infecting an individual through nasal inhalation. Here, we note that Chen teaches that its composition is applied "in and around the site of intended treatment," including "treatment on the nasal mucous membrane." Ex. 1005 ¶ 110. Chen teaches the use of "[a]ny methods appropriate for . . . administering a pharmaceutical composition to a subject" that "generally cause the formulation to coat and remain in contact with those membranes or skin surfaces for a period of time, like a chemical barrier at those sites." *Id.* ¶112. Chen also provides an exemplary formulation that is "a viscous gel, which provides excellent muco-adhesiveness by forming a physical barrier over the mucous membrane for a prolonged period of time." *Id.* ¶ 121 (Example 1).

We recognize Patent Owner's contention that the Specification of the '802 patent provides examples of effective formulations according to the disclosed invention that would indicate the need for "a purposeful adjustment of the positive charge so as to effectively create an electrostatic field surrounding the user's nose and face that would attract airborne particles that float in the vicinity of the nostrils." PO Sur-Reply 7–8 (citing Ex. 1001, 5:55–10:5 (Tables 1–10)). We note here, however, that the claims do not recite language specifying a particular result nor require a particular amount or degree of electrostatic attraction of the particulate matter to the thin film. Nor do the claims require specific concentrations of compounds in the formulation.

On this record, Petitioner has demonstrated by a preponderance of evidence that Chen's positively charged composition would necessarily, and thus inherently, electrostatically attract negatively charged viral particles for the purposes of "ameliorating viral infection" by "inhibit[ing] or reduc[ing] the contact between the viral particles and the subject," the same purpose of the challenged claims as articulated by Patent Owner. Ex. 1005 ¶¶ 41, 53, 109; PO Resp. 10; *see also id.* ¶¶ 67, 126 (Example 3, identifying formulations most effective in inhibiting the adenovirus); Pet. 19; Pet. Reply 15–16; Ex. 1002 ¶¶ 126–130.[4] Patent Owner does not otherwise direct us to

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[4] A reference may inherently anticipate if a claim limitation that is not expressly disclosed "is necessarily present, or inherent, in the single anticipating reference." *Verizon Servs. Corp. v. Cox Fibernet Va., Inc.*, 602 F.3d 1325, 1337 (Fed. Cir. 2010). Thus, the "very essence of inherency is

any evidence that electrostatically attracting harmful particulate matter requires a concentration of a cationic agent greater than what is disclosed in Chen. *See*, *e.g.*, Ex. 1005 ¶ 121 (Example 1).

After considering the parties' arguments and evidence, we are persuaded that Chen discloses a formulation that "electrostatically attracts the particulate matter to the thin film" as required by claim 2.

> c) *Claim Element 2b: "holds the particulate matter in place by adjusting the adhesion of the thin film to permit said thin film to stick to the skin or tissue and by adjusting the cohesion of the formulation to provide adequate impermeability to the thin film"*

Petitioner asserts that Chen discloses the "HOLD" element of claim 2.[5] Pet. 20–21. According to Petitioner and Dr. Park, Chen's formulation functions in this way because Chen's cationic agent in the formulation attracts and holds particulate matter, and the thin film comprising the cationic agent and formulated as a gel, lotion, cream, or ointment holds the particulate matter to the film. *Id*. (citing Ex. 1005 ¶ 38, Ex. 1002 ¶¶ 77, 134).

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that one of ordinary skill in the art would recognize that a reference *unavoidably* teaches the property in question." *Agilent Technologies, Inc. v. Affymetrix, Inc.*, 567 F.3d 1366, 1383 (Fed. Cir. 2009) (emphasis added).

[5] Claims 1 and 2 recite a formulation capable of holding particulate matter in place "by adjusting the adhesion of the thin film to permit said thin film to stick to the skin or tissue and by adjusting the cohesion of the formulation to provide adequate impermeability to the thin film." As noted above, the parties refer to this element of claims 1 and 2 as the "HOLD" function. *See*, *e.g*., PO Resp. 11; Pet. Reply, 17.

Petitioner also asserts that Chen discloses adjusting the adhesion and cohesion of the formulation by exemplifying different formulations that "vary the concentration of ingredients." *Id*. at 21 (citing Ex. 1005 ¶ 97). For example, Petitioner compares Chen's compositions 12 and 14, which reflect an adjustment of the cationic polymer from 1.5 wt.% to 0.15 wt.%. *Id*. According to Petitioner and Dr. Park, "[c]hanging the amount of cationic polymer in a composition will change the adhesion and cohesion of a formulation applied as a thin film." *Id*. (citing Ex. 1002 ¶ 134).

Patent Owner opposes, contending that Chen was unaware that modifying cohesion would improve the effectiveness of the disclosed formulation. PO Resp. 11–17. In particular, Patent Owner suggests that while Chen may have been interested in modifying the viscosity of the disclosed formulations, modification of the viscosity does not necessarily result in a change to a formulation's cohesion. *Id.* at 13. Specifically, Patent Owner contends that,

> while viscosity modifiers can render a formulation more viscous, they have no effect on cohesion. Cohesion is the ability of molecules of the same material to bond together. While increased cohesion will increase viscosity, the converse is not necessarily true. Guar gum is often used as a viscosity modifier to increase the flow of oil through cylindrical pipes, inclusion of guar does not make the oil less cohesive. Mud or clay in a fluid increases the viscosity of the fluid, but it does not increase the bonding strength of the fluid molecules. To increase the cohesiveness of a polymer gel, one increases the length of the molecular chain and the molecular weight of the polymer. Permeability depends upon cohesion rather than viscosity.

Id. at 13–14 (citing Ex. 1005 ¶ 115 ("By forming a viscous membrane over nasal mucosa and by deactivating viruses on contact, the formulation of the present invention could provide an effective means to prevent the virus from spreading.”).

Patent Owner further contends that

> While adequate permeability refers to “inhibiting harmful particles from penetrating the thin film and contacting the skin or tissue of an individual’s nasal passages,” it does not provide for “holding the particulate matter in place” as recited in the claims of the ’802 Patent. This is a critical element of the claims of the ’802 Patent, and it is not present in Chen.

PO Resp. 16 (emphasis omitted). Patent Owner contends that

> There is no indication whether there has been FDA approval for any human trials of Chen’s compositions. If the compositions were not tested on human beings, no knowledge could exist as to whether virus particles are dislodged from his composition when subjected to air flow resulting from nasal breathing. . . . Chen is silent regarding holding the virus particles in place. Chen is silent regarding preventing virus particles from becoming free agents. Chen does not blow air over his anti-viral compositions to simulate nasal breathing and to determine whether live viruses can escape into the air stream. He was unaware of the problem.

Id. at 12–13. Thus, according to Patent Owner, without the knowledge of the problem to be solved, a POSITA would not have sought to adjust the cohesion of the formulation to provide adequate impermeability to the thin film as required by claim 2. *Id.* at 17.

Having considered the parties’ positions and evidence of record, we determine that Chen discloses a formulation having the “HOLD” element of claim 2. To begin, Chen discloses that its composition is applied “in and around the site of intended treatment,” including “treatment on the nasal mucous membrane.” Ex. 1005 ¶ 110. Chen teaches the use of “[a]ny methods appropriate for . . . administering a pharmaceutical composition to a subject” that “generally cause the formulation to coat and remain in contact with those membranes or skin surfaces for a period of time, like a chemical barrier at those sites." *Id*. ¶ 112. Chen also provides an exemplary formulation that is "a viscous gel, which provides excellent muco-adhesiveness by forming a physical barrier over the mucous membrane for a prolonged period of time." *Id*. ¶ 121, Example 1.

Additionally, Chen discloses that cationic surfactants are useful for its antiviral compositions, including "those in the form of . . . quaternary ammonium salts," e.g., "cetyl trimethyl ammonium chloride . . . and alkylbenzyl dimethyl ammonium chloride (such as benzalkonium or benzethonium type preservative, disinfectant and fungicide)." Ex. 1005 ¶¶ 82–84; *see also id*. at ¶¶ 85–86 (Chen disclosing that benzalkonium chloride (i.e., alkyldimethyl (phenylmethyl) ammonium chloride) is "a surface-active germicide for many pathogenic nonsporulating bacteria and fungi."). As noted by Petitioner,

> Chen discloses adjusting the adhesion and cohesion of the formulation to provide adequate impermeability to the thin film . . . . Chen discloses multiple examples showing different formulations that "vary the concentration of ingredients." (Ex. 1005 [0097].) For instance, cationic poly(dimethyl-aminoethylmethacrylate), may be adjusted from 1.5 wt.% in Antiviral Composition 12 to 0.15 wt.% in Antiviral Composition 14.

Pet. 21; *see also* Ex. 1005 ¶ 92 ("the cationic polymer component is present in an amount (in chloride salt equivalent) from about 0.001% to about 2%"); Ex. 1002 ¶ 183 ("Chen discloses poly(methyl methacrylate-b-N-methyl-4-vinyl pyridinium iodide[, which] is a polymeric quaternary ammonium compound."). Here, we credit the testimony of Dr. Park that "[c]hanging the amount of cationic polymer in a composition will change the adhesion and cohesion of a formulation applied as a thin film." Ex. 1002 ¶ 134.

> Furthermore, in view of the above, we agree with Petitioner that
>
> Chen's disclosed film necessarily performs Claim Element b), and whether Chen recognized this function or not is irrelevant. (Ex. 1002 ¶¶ 165-169 (claim 1), 131-140 (claim 2).) Chen's formulation necessarily performs "holding the particulate matter in place by adjusting the adhesion of the thin film to permit said thin film to stick to the skin or tissue and by adjusting the cohesion of the formulation to provide adequate impermeability to the thin film." *Id*. Thus, this limitation is met.

Pet. Reply 18.

We are not persuaded by Patent Owner's argument that, without the purposeful adjustment of the concentration of cationic ingredients, Chen does not disclose a composition capable of achieving "adequate impermeability" as required by claim 2. *See* PO Resp. 12 ("The HOLD element attribute of the formulation cannot be achieved accidentally."). Similar to our discussion with regard to claim element 2a, the claims do not recite language specifying a particular outcome (e.g., increasing cohesion). Nor do the claims require specific concentrations of compounds in the formulation or even recite a direction for which to vary the relative concentrations of compounds.

On this record, we determine that Petitioner has demonstrated by a preponderance of evidence that Chen's composition is capable of achieving "adequate impermeability" as recited in claim 2. That is, Chen's positively charged composition would necessarily, and thus inherently, provide "adequate impermeability" by creating a thin film capable of "forming a physical barrier over the mucous membrane for a prolonged period of time" for the purposes of "ameliorating viral infection" by "inhibit[ing] or reduc[ing] the contact between the viral particles and the subject." Ex. 1005 ¶¶ 41, 53, 109, 121. In other words, "[b]y forming a viscous membrane over nasal mucous and by deactivating viruses on contact," Chen's formulation "provide[s] an effective means to prevent the virus from spreading," fulfilling the HOLD requirement of the claims. *Id.* ¶ 115; *see also* Pet Reply 17 ("Patent Owner has not provided any evidence as to why Chen's formulation would not provide the 'HOLD' function."). Furthermore, adjustments to the adhesion or cohesion of Chen's formulations are made by varying the concentrations of components to achieve the desired effect, which Chen discloses. Pet. 21; Ex. 1005 ¶¶ 23, 55, 84, 87; Ex. 1002 ¶¶ 131–138.

We are also not persuaded by Patent Owner's argument that Chen cannot anticipate the subject matter of claim 2 as Chen was unaware of the problems addressed by adjusting the cohesion of the disclosed compositions. PO Resp. 12–13. Chen discloses the use of quaternary ammonium salts and also discloses effective ranges for adjusting the concentration of cationic surfactants such as quaternary ammonium salts in its disclosed compositions. Ex. 1005 ¶¶ 84, 93. That disclosure alone is sufficient to demonstrate anticipation because Chen's composition falls within the scope of claim 2, as discussed above. *Atlas Powder Co. v. Ireco, Inc.*, 190 F.3d 1342, 1346 (Fed. Cir. 1999) ("Anticipation of a patent claim requires a finding that the claim at issue 'reads on' a prior art reference.").

Even if Chen failed to recognize "cohesion" as a result-effective variable, it is well established under Federal Circuit precedent that inherent properties of a chemical composition are present regardless of whether they are known or intended. *Id.* at 1347 ("[T]he discovery of a previously unappreciated property of a prior art composition, or of a scientific explanation for the prior art's functioning, does not render the old composition patentably new to the discoverer.”); *see also Mehl/Biophile Intern. Corp. v. Milgraum*, 192 F.3d 1362, 1366 (1999) (“Where, as here, the result is a necessary consequence of what was deliberately intended, it is of no import that the articles’ authors did not appreciate the results.”); *see also* Pet. Reply 17–18. Claim 2 merely requires adjusting adhesion and cohesion, and we agree with Petitioner that this HOLD element of claim 2 is met by Chen’s disclosure of varying the concentration of the cationic components in Chen’s composition, especially compositions comprising a cationic surfactant, the class of compounds that Chen discloses as including benzalkonium chloride and other quaternary ammonium salts. Pet. 21; Ex. 1005 ¶¶ 23, 55, 84, 87; Ex. 1002 ¶¶ 131–138.

Thus, after considering the parties’ arguments and evidence, we determine that Chen discloses the “HOLD” element of claim 2. In reaching that determination, and in addition to our discussion above, we credit Dr. Park’s testimony that a person of skill in the art would have understood that Chen’s composition was formulated so that particulate matter attracted to Chen’s formulation would remain in place on the film and not penetrate through to the skin because the formulation cohesion provided adequate impermeability. Ex. 1002 ¶¶ 131–139. Dr. Park also testifies that adjustments to the cohesive properties of Chen’s compositions are achieved by changing the amount of cationic polymer in the composition, which will also adjust the viscosity of the composition. *Id.* ¶¶ 134–135, 137. Finally, we credit Dr. Park’s testimony that a person of ordinary skill in the art would have understood that Chen’s compositions would have adequate impermeability to prevent viruses or other harmful particulate matter attracted to the film from reaching the skin, as it would have been apparent that achieving Chen's antiviral activity would require this function. *Id*. ¶ 138.

d) Claim Element 2c: "inactivates the particulate matter and renders said particulate matter harmless"

Petitioner asserts that Chen discloses the "KILL" element of claim 2.[6] Pet. 22. Petitioner asserts that Chen's formulation "inactivates the particulate matter and renders said particulate matter harmless," as required by claim 2, because those actions are achieved by the biocidic agent, BAC, in Chen's formulation. *Id*. (citing Ex. 1002 ¶ 143). Alternatively, Petitioner asserts that a skilled artisan would have understood that the time of the invention that Chen's formulations comprising a biocidic agent inherently inactivate the particulate matter and render that matter harmless. *Id*.

Patent Owner does not raise any arguments regarding the preamble of claim 2. *See generally* PO Resp. The arguments and evidence that Petitioner cites set forth above persuade us that Chen teaches this element of claim 2.

e) Conclusion on Claim 2

Upon considering the full trial record, we find by a preponderance of the evidence that Chen discloses each element of claim 2. We, therefore, conclude that claim 2 is anticipated by Chen.

[6] Claims 1 and 2 recite a formulation capable of inactivating the "particulate matter by adding at least one ingredient that would render said particulate matter harmless." As noted above, the parties refer to this element of claims 1 and 2 as the "KILL" function. *See*, *e.g*., PO Sur-Reply 11 n.11.

2. Dependent Claim 3

Claim 3 depends from claim 2 and further recites "wherein the at least one cationic agent is a polymeric quaternary ammonium compound." As discussed above, we find that Chen expressly discloses the use of a polymeric quaternary ammonium compound, i.e., poly(methyl methacrylate-b-N-methyl-4-vinyl pyridinium iodide) in its compositions. *See* Pet. 22–23 (citing Ex. 1005 ¶ 78; Ex. 1002 ¶ 149).

Patent Owner does not argue that the additional limitation of claim 3 is not taught by Chen. *See generally* PO Resp.

Upon considering the full trial record, we find by a preponderance of the evidence that Chen discloses the subject matter of claim 3. *See* Pet. 22–23; Ex. 1005 ¶ 78; Ex. 1002 ¶ 149. We, therefore, conclude that claim 3 is anticipated by Chen.

3. Independent Claim 1

For claim 1, Petitioner reiterates its discussion of claim 2 to support its assertion that Chen anticipates or renders obvious claim 1. *See* Pet. 23–26. Additionally, for this method claim, Petitioner relies on Chen's disclosure of "a method for ameliorating viral infection comprising administering to a subject in need thereof the composition described herein." *Id*. at 24 (quoting Ex. 1005 ¶ 109). Patent Owner does not present arguments in addition to those we have discussed concerning claim 2 against Petitioner's challenge to claim 1 as being anticipated by Chen. *See generally* PO Resp.

For the same reasons discussed above with regard to claim 2, we find by a preponderance of the evidence that Chen discloses the subject matter of claim 1. *See* Pet. 23–26; Ex. 1002 ¶¶ 152–177. We, therefore, conclude that claim 1 is anticipated by Chen.

4. *Independent Claim 8*

Claim 8, unlike claims 1 and 2, does not include the language regarding adjustment of the adhesion and cohesion of the thin film. The formulation of claim 8 is defined as having at least one biocidic agent and at least one quaternary thickener. For claim 8, Petitioner reiterates its discussion of claim 2 to support its assertion that Chen discloses each element of claim 8. Pet. 27–28. Additionally, Petitioner asserts that the polymeric quaternary ammonium compound in Chen's compositions "would increase the thickness of the compositions," and therefore function as a "quaternary thickener[]." Pet. 27 (citing Ex. 1005 ¶ 78, Ex. 1002 ¶¶ 183–184). Patent Owner does not present arguments in addition to those we have discussed concerning claim 2 directed at the patentability of claim 8. *See generally* PO Resp.

Having considered the parties' positions and evidence of record, summarized above, we find by a preponderance of the evidence that Chen discloses the subject matter of claim 8. Pet. 27; Ex. 1002 ¶¶ 183–184. We, therefore, conclude that claim 8 is anticipated by Chen.

5. *Conclusion*

The preponderance of the evidence supports Petitioner's argument that Chen discloses each and every element of the challenged claims. We are not persuaded otherwise by Patent Owner's arguments. Accordingly, we determine that claims 1–3 and 8 are anticipated by Chen.

C. Ground 2: Asserted Obviousness over Chen

Petitioner argues that the same claims challenged under Ground 1 as anticipated by Chen would also have been obvious over Chen. Pet. 16; *see also* Pet. 20 (Petitioner asserting that it would have been obvious to a skilled artisan that Chen's formulations comprising cationic polymers "would inherently electrostatically attract the particulate matter."); *id.* at 21 (Petitioner asserting that it would have been obvious to modify Chen's formulation "so that the formulations would hold the particulate matter and would have adequate impermeability to prevent the viruses from reaching the skin, tissue, or mucosa of the nasal passages, with a reasonable expectation of success.").

Patent Owner contends a POSITA would only envisage the HOLD function of the challenged claims "if he or she recognized the necessity of that function. There is no evidence that . . . would occur unless the POSITA was aware" of the problem associated with dislodged viruses. PO Resp. 19–20.

Because "anticipation is the epitome of obviousness," we are persuaded that claims 1–3 and 8 that we have determined are anticipated by Chen would have been obvious over Chen for the same reasons as discussed above. *In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002). (*See* Pet. Reply 21.) Accordingly, the preponderance of the evidence supports Petitioner's challenge of claims 1–3 and 8 as obvious over Chen.

Patent Owner also presents objective evidence of non-obviousness that it asserts demonstrates the non-obviousness of the claimed methods. PO Resp. 21–22. The evidence purportedly shows commercial success of the claimed formulations. *Id.* Because we determine, as discussed above, that the challenged claims are anticipated by Chen and, a fortiori, obvious as well, Patent Owner's objective evidence of non-obviousness is not persuasive of the patentability of claims 1–3 and 8 under either the anticipation or obviousness challenge. *See Cohesive Tech., Inc. v. Waters Corp.*, 543 F.3d 1351, 1364 (Fed. Cir. 2008) ("secondary considerations are not an element of a claim of anticipation.").

Accordingly, we also determine that Petitioner has demonstrated by a preponderance of the evidence that claims 1–3 and 8 would have been obvious as well as anticipated by Chen. We are not persuaded to the contrary by Patent Owner's arguments, and find Patent Owner's evidence of second secondary considerations to be inapposite to this patentability determination because the claims are anticipated by Chen as well.

D. Grounds 3 and 4: Asserted Unpatentability in view of Baker

Petitioner asserts that Baker anticipates claims 1–3 and 8. Pet. 28–40. To support its contention, Petitioner provides a detailed claim analysis addressing how each element of 1–3 and 8 is disclosed by Baker. *Id.* Petitioner supports this application of Baker with Dr. Park's testimony. Ex. 1003 ¶¶ 187–273. Alternatively, Petitioner asserts that "[i]n the event that one or more of claims 1-3 or 8 are not found to be anticipated by [Baker], claims 1-3 and 8 would nevertheless have been obvious over [Baker]." *Id*. at 40. Patent Owner disagrees.[7]

[7] The Patent Owner Response does not separately dispute Petitioner's challenges relying on Baker, Wahi'005, and/or Rolf (i.e. Grounds 3–9). Rather, Patent Owner attempts to incorporate by reference its arguments presented in Patent Owner's Preliminary Response with regard to those grounds, which is not proper. 37 C.F.R. § 42.6(a)(3); *see also* Paper 8, 9 ("[A]ny arguments not raised in the response may be deemed waived.").

Having considered the parties' positions and evidence of record, we determine that Petitioner has established by a preponderance of evidence that claims 1–3 and 8 are anticipated by Baker for the reasons stated in the Petition as further supported by the testimony of Dr. Park. Pet. 28–40. Briefly, we find that Baker discloses "compositions and methods for decreasing the infectivity . . . associated with a variety of pathogenic organisms and viruses." Ex. 1006, Abst. Baker provides a composition applied to the surface of the skin and mucosal cells and tissues (including nasal mucosa) comprising a quaternary ammonium compound, for example N,N-Dimethyl-2-hydroxypropyl ammonium chloride polymer. *Id.* at 5:66–67; 7:1–2; 13:24–26; 36:12–14. Baker also discloses a composition that "can take the form of a topical, cream, lotion, or gel and can include emulsifiers, gelling agents." Pet. 33 (citing Ex. 1006 36:31–34). Baker's composition further comprises at least one biocidic agent (e.g. BAC or TWEEN 60). Ex. 1006, 25:28–29, 25:63, 30:39.

Furthermore, we agree with Petitioner that a POSITA would understand that Baker's compositions are applied as a thin film to skin or tissue of nasal passages of an individual. Pet. 32 (citing Ex. 1006 36:12–16; Ex. 1002 ¶ 200 ("A POSITA would understand that these compositions are applied as a thin film.")). We also agree with Petitioner that

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Notwithstanding, we acknowledge Patent Owner's general argument that none of Petitioner's references cited in Grounds 3–9 anticipates the challenged claims, and, in particular, fail to disclose the "HOLD" element of the challenged claims. *See* PO Sur-Reply 3 ("The substance of the arguments related to Grounds [3–9] is implicitly presented by the broader discussion of the '802 Patent's distinct features, particularly of the HOLD function, which overcomes all prior art submitted by the Petitioner.").

> Baker's thin films can include cationic agents such as cationic halogen containing compounds (Ex. 1006 17:5-41) and quaternary ammonium compounds (*Id*. 5:64-7:12) including N,N-Dimethyl-2-hydroxypropylammonium chloride polymer. (*Id*. 7:1-2.) Cationic agents electrostatically attract particulate matter and therefore hold the particulate matter. (Ex. 1002 ¶226.)

Pet. 33. Specific to the "HOLD" function of the challenged claims, Baker discloses adjusting the adhesion and cohesion of the formulation to provide adequate impermeability because "Baker shows multiple formulations that 'vary[] the concentration of ingredients.'" Pet. 33 (citing Ex. 1006, Figs. 29A–29E). Accordingly, in view of the above, the preponderance of evidence of record supports our determination that Baker anticipates the challenged claims.

With regard to Petitioner's obviousness challenge, we determine that the preponderance of the evidence supports Petitioner's challenges of claims 1–3 and 8 as obvious over Baker. Because "anticipation is the epitome of obviousness," we are persuaded that the claims Petitioner challenges as being anticipated by the Baker would have been obvious over Baker for the reasons discussed above. *McDaniel*, 293 F.3d at 1385. Furthermore, because we determine that the subject matter of the challenged claims is anticipated by Baker, Patent Owner's objective evidence of non-obviousness is not persuasive of the patentability of claims 1–3 and 8. *See Cohesive Tech.*, 543 F.3d at 1364 ("secondary considerations are not an element of a claim of anticipation.").

Accordingly, we also determine that Petitioner has demonstrated by a preponderance of the evidence that claims 1–3 and 8 would have been obvious over Baker. We are not persuaded to the contrary by Patent Owner's arguments or evidence of second secondary considerations.

E. Grounds 5 and 6: Asserted Unpatentability in view of Wahi '005

Petitioner asserts that Wahi '005 anticipates claims 1–3 and 8. Pet. 40–50. To support its contention, Petitioner provides a detailed claim analysis addressing how each element of 1–3 and 8 is disclosed by Wahi '005. *Id.* Petitioner supports this interpretation of Wahi '005 with Dr. Park's testimony. Ex. 1003 ¶¶ 274–354. Alternatively, Petitioner asserts that "[i]n the event that one or more of claims 1-3 or 8 are not found to be anticipated by Wahi '005, claims 1-3 and 8 would nevertheless have been obvious over Wahi '005." *Id*. at 40. Patent Owner disagrees. PO Sur-Reply 3.

Having considered the parties' positions and evidence of record, we determine that Petitioner has established by a preponderance of evidence that claims 1–3 and 8 are anticipated by Wahi '005 for the reasons stated in the Petition, as further supported by the testimony of Dr. Park. Pet. 40–50. Briefly, we find that Wahi '005 discloses "products for restricting the flow of airborne contaminants into a nasal passage by creating an electrostatic field of increased charge in an area about the nasal passage[, which] prevents or reduces the inflow of airborne contaminants to the nasal passage." Ex. 1007, 1:21–26. Wahi '005 discloses that the "nasal application product" may be an ointment, paste, cream, or gel, which are formulations applied to the skin or tissue of nasal passages in a thin film. *Id.* at 2:22–26; Ex. 1002 ¶ 281.

Wahi '005 discloses electrostatic polymers such as poly (dimethyl diallyl ammonium chloride) to increase the electrostatic charge density of its formulations. Ex. 1007, 3:31–40. For example, Wahi '005 discloses compositions comprising Celquat SC 240C (polyquaternary ammonium cellulose), *id.* at 4:32–33, and Agequat 400 (a poly(dimethyl diallyl ammonium chloride)), *id.* at 3:34–35, both of which are cationic agents as well as biocidic agents (Ex. 1002 ¶¶ 287, 291). Specific to the "HOLD" function of the challenged claims, we agree with Petitioner that

> Wahi '005 discloses multiple examples showing different formulations that "vary the concentration of ingredients." (Ex. 1007 3:45-4:63.) For instance, Example 1 has 5 wt. % of Agequat 400 and Example 4 has 11 wt.%. Agequat is a polymer. (Ex. 1002 ¶304.) Changing the amount of cationic polymer in a formulation will change the adhesion and cohesion of a formulation applied as a thin film. (*Id.* ¶305.)

Pet. 44. We find that modifying the concentration of the cationic polymer in a formulation adjusts the adhesion and cohesion of a formulation. Ex. 1002 ¶ 305; *see also* Ex. 1007, 3:32–35 ("a significant increase in electrostatic charge density is achieved with the present invention by use of at least 10% by weight of poly (dimethyl diallyl ammonium chloride)."). Accordingly, in view of the above, the preponderance of evidence of record supports our determination that Wahi '005 anticipates the challenged claims.

With regard to Petitioner's obviousness challenge, we determine that the preponderance of the evidence supports Petitioner's challenges of claims 1–3 and 8 as obvious over Wahi '005. Because "anticipation is the epitome of obviousness," we are persuaded that the claims Petitioner challenges as being anticipated by the Wahi '005 would have been obvious over Wahi '005 for the reasons discussed above. *McDaniel*, 293 F.3d at 1385. Furthermore, because we determine that the subject matter of challenged claims is anticipated by Wahi '005, Patent Owner's objective evidence of non-obviousness is not persuasive of the patentability of claims 1–3 and 8.

See Cohesive Tech., 543 F.3d at 1364 ("secondary considerations are not an element of a claim of anticipation.").

Accordingly, we also determine that Petitioner has demonstrated by a preponderance of the evidence that claims 1–3 and 8 would have been obvious over Wahi '005. We are not persuaded to the contrary by Patent Owner's arguments or evidence of second secondary considerations.

F. Ground 7: Asserted Anticipation by Rolf

Petitioner asserts that Rolf anticipates claims 1, 2, and 8. Pet. 50–60. To support its contention, Petitioner provides a detailed claim analysis addressing how each element of 1, 2, and 8 is disclosed by Rolf. *Id.* Petitioner supports this interpretation of Rolf with Dr. Park's testimony. Ex. 1003 ¶¶ 356–430. Alternatively, Petitioner asserts that "[i]n the event that one or more of claims 1, 2, and 8 are not found to be anticipated by Rolf, claims 1, 2, and 8 would nevertheless have been obvious over Rolf." *Id*. at 50. Patent Owner disagrees. PO Sur-Reply 3.

Having considered the parties' positions and evidence of record, we determine that Petitioner has established by a preponderance of evidence that claims 1, 2, and 8 are anticipated by Rolf for the reasons stated in the Petition, supported by the testimony of Dr. Park. Pet. 50–60. Briefly, we find that Rolf discloses "a method for preventing a respiratory infection in a mammal at risk thereof" comprising placing an adhesive skin patch "in the vicinity of the nasal passageway." Ex. 1008, Abst. The patch holds essential oils that deactivate respiratory pathogens upon contact with the essential oil. *Id.* ¶ 20. The patch uses an adhesive backing, or "ointment or gel" on the backing of the patch, that contains Quat-15, a cationic agent, and is therefore capable of electrostatically attracting the particulate matter. *Id.*

at ¶¶ 42, 44, 107, 149; Ex. 1002 ¶¶ 378–381. Specific to the "HOLD" function of the challenged claims, we agree with Petitioner that

> Rolf discloses 16 examples that "vary[] the concentration of ingredients," and "thereby adjust[] the adhesion and cohesion of the thin film." (Ex. 1008 [0268]-[0283].) Each of the Examples states that the formulation may comprise a percentage range of an ingredient and then lists a "typical" (i.e., preferred) percentage. For instance, in Example 6 in paragraph [0273] has a range percentage of 10% to 50% polyvinyl alcohol; Example 8 in paragraph [0275] has a range percentage of 10% to 40%. "Varying the concentration" of polyvinyl alcohol in a formulation by up to 40% will "adjust[] the adhesion and cohesion of the thin film." (Ex. 1002 ¶392.)

Pet. 54–55. Furthermore, we find that modifying the concentration of the cationic polymer in a formulation adjusts the adhesion and cohesion of a formulation. Ex. 1008 ¶ 149 ("quat-15 can be employed in about 0.01 wt. % to about 1.5 wt. % of the formulation 5"). Accordingly, in view of the above, the preponderance of evidence of record supports our determination that Rolf anticipates the challenged claims.

G. Grounds 8 and 9: Asserted Obviousness in view of Rolf or the combination of Rolf and Wahi '005

With regard to Petitioner's obviousness challenges that rely on Rolf, we determine that the preponderance of the evidence supports Petitioner's challenges of claims 1, 2, and 8 as being obvious over Rolf. Because "anticipation is the epitome of obviousness," we are persuaded that the claims Petitioner challenges as being anticipated by the Rolf would have been obvious over Rolf for the reasons discussed above. *McDaniel*, 293 F.3d at 1385. Furthermore, because we determine that the subject matter of the challenged claims is anticipated by Rolf, Patent Owner's objective evidence of non-obviousness is not persuasive of the patentability of claims

1, 2, and 8. *See Cohesive Tech.*, 543 F.3d at 1364 ("secondary considerations are not an element of a claim of anticipation.").

Accordingly, we also determine that Petitioner has demonstrated by a preponderance of the evidence that claims 1, 2, and 8 would have been obvious in view of Rolf or the combination of Rolf and Wahi '005. We are not persuaded to the contrary by Patent Owner's arguments or evidence of second secondary considerations.

IV. PETITIONER'S MOTION TO EXCLUDE

Petitioner moves to exclude Exhibits 2003–2007. Paper 22, 1. Because our decision does not rely on any of the challenged exhibits, we dismiss Petitioner's Motion to Exclude Evidence as moot.

V. PATENT OWNER'S MOTION TO STRIKE

Petitioner moves to strike Exhibits 2003–2007. Paper 19, 1–2. Because we do not rely in our analysis on these exhibits in a manner adverse to Petitioner, we dismiss Petitioner's Motion to Strike as moot as to these exhibits.

VI. CONCLUSION[8]

Based on the fully developed trial record, we determine that Petitioner has shown by a preponderance of the evidence that the challenged claims are unpatentable as summarized below.

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[8] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent

Claims	35 U.S.C. §	Reference(s)/ Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–3, 8	102(b)	Chen	1–3, 8	
1–3, 8	103	Chen	1–3, 8	
1–3, 8	102(b)	Baker	1–3, 8	
1–3, 8	103	Baker	1–3, 8	
1–3, 8	102(b)	Wahi ’005	1–3, 8	
1–3, 8	103	Wahi ’005	1–3, 8	
1, 2, 8	102(b)	Rolf	1, 2, 8	
1, 2, 8	103	Rolf	1, 2, 8	
2	103	Rolf, Wahi ’005	2	
Overall Outcome			1–3, 8	

VII. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has proved by a preponderance of the evidence that claims 1–3 and 8 are unpatentable;

FURTHER ORDERED that Petitioner's Motion to Exclude is *dismissed* as moot;

FURTHER ORDERED that Petitioner's Motion to Strike is *dismissed* as moot; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

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